

HOMEWORK 1

1. Let $a_1, a_2, \dots, a_n$ be elements of a group G . Define the product of the a_i 's by induction:
 $a_1 a_2 \cdots a_n = (a_1 a_2 \cdots a_{n-1}) a_n$.

(a) Prove that

$$a_1 a_2 \cdots a_n b_1 b_2 \cdots b_m = (a_1 a_2 \cdots a_n)(b_1 b_2 \cdots b_m).$$

(b) Prove that $a_1 a_2 \cdots a_n$ is equal to the product of the a_i 's with the parentheses inserted arbitrarily.

2. (a) Prove that for every integer $n > 0$, the set of all complex n th roots of unity is a group with respect to complex multiplication. Show that this group is cyclic.

(b) Prove that if G is a cyclic group of order n and k divides n , then G contains exactly one subgroup of order k .

3. (a) Show that if K and N are two finite subgroups of a group G of relatively prime orders, then $K \cap N = \{e\}$.

(b) Show that if a group G has only finite number of subgroups, then G is finite.

4. Prove that a group G is cyclic if and only if there is an element $a \in G$ with $\text{ord}(a) = |G|$.

5. (a) Show that if $a^2 = e$ for all elements a of a group G , then G is abelian.

(b) Prove that if G is a finite group of even order, then G contains an element a such that $a^2 = e$ and $a \neq e$.

(c) Show that every subgroup of index 2 is normal.

6. Find all groups (up to isomorphism) of order ≤ 5 . What is the smallest order of a non-cyclic group?

7. Find all subgroups of the symmetric group S_3 . Determine normal subgroups.

8. Let n be a natural number. Show that the map

$$f : \mathbb{Q}/\mathbb{Z} \rightarrow \mathbb{Q}/\mathbb{Z}, \quad f(a + \mathbb{Z}) = na + \mathbb{Z}$$

is a well defined homomorphism. Find $\text{Ker}(f)$ and $\text{Im}(f)$.

9. (a) Show that $\mathbb{Z}/6\mathbb{Z} \simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$.

(b) Prove that $n\mathbb{Z}/mn\mathbb{Z} \simeq \mathbb{Z}/m\mathbb{Z}$.

10. Let $f : G \rightarrow H$ be a surjective group homomorphism and let H' be a subgroup in H .

(a) Show that $f^{-1}(H')$ is a subgroup of G .

(b) Prove that the assignment $H' \mapsto f^{-1}(H')$ yields a bijection between the set of all subgroups of H and the set of all subgroups of G that contain $\text{Ker}(f)$.

(c) Prove that a subgroup H' is normal in H if and only if $f^{-1}(H')$ is normal in G . Show that $G/f^{-1}(H') \simeq H/H'$.